

Glass Box Trading

AI proposes; deterministic gates dispose.

- Result at 2026-09-03T20:00:14.787Z: \$583.59 (0.58%) on a \$100k paper account
- Every decision — trade or no trade — is public and broker-reconciled
- Demo: <https://glass-box-trading.vercel.app> · Repo: <https://github.com/fradzano/glass-box-trading>

The auditability problem

- "Autonomous trading agent" claims are easy to make, hard to verify
- Target user: a trader or allocator who wants unattended execution but distrusts opaque bot claims
- This entry publishes the whole decision trail — proposals, vetoes, and fills alike — not just the wins

The decision-to-fill golden path

1. Open the dashboard, no login — see account state and control model
2. Pick one decision cycle — market context, candidate, rationale
3. See the full gate vector, including at least one veto
4. Follow one approved intent to its order, fill, and P&L contribution
5. Open the public source at the pure decision core and its tests

"AI proposes; deterministic gates dispose"

- LLM analyst proposes candidates from read-only Alpaca market data (MCP)
- Analyst output is schema-validated and whitelist-constrained — never a free-form order
- A pure, tested decision core owns every risk gate; time/config/account state are parameters, not ambient calls
- Executor places only core-approved actions through the Alpaca REST API (multi-leg limit orders, client order id as idempotency key) — the LLM has no code path to an order

Options strategy and declared variance allocation

- Two-sleeve barbell, declared up front, not fit after the fact:
 - Reserve ~80% — cash, idle
 - Income ~12% (max-loss basis) — defined-risk credit spreads / iron condors
 - Convex ~8% (premium basis) — debit spreads / long options
- Defined-risk only, everywhere — no naked short options in the strategy space

The cycle and its gates, in order

1. Phase 0: reconcile the journal against broker truth; anything foreign halts
2. Fresh broker snapshot and quotes; the analyst proposes candidates (read-only MCP)
3. The pure core prices each candidate from its own quotes and runs
G1 defined risk only · **G2** sleeve budgets · **G3** max loss per position ·
G4 per-underlying concentration · **G5** liquidity · **G6** session and tradability · **G7** idempotency · **G8** schema and whitelist
4. Executor: limit order, revalidated against a fresh broker read, INTENT before, OUTCOME after
 - Lifecycle and failure gates run every cycle: **G9** expiry eviction, **G10** reconciliation, **G11** deadline flatten, **G12** single instance and halt, **G13** drawdown kill-switch, **G14** dead-man watchdog
 - Worst case is bounded by design, not by monitoring diligence

Broker-reconciled P&L and sleeve attribution

- Account PA376WIK2ATL, journal revision sha256:7b82959a344a7c7e
- Start equity \$100,000.00 · Equity at cutoff \$100,583.59 · P&L \$583.59 (0.58%) at 2026-09-03T20:00:14.787Z
- Realized \$586.00 · Unrealized \$0.00 · Unattributed -\$2.41 — a fee-shaped residual the journal cannot explain, shown and never assigned to a sleeve
- Realized only, by sleeve: income \$230.00 (5 filled spreads) · convex \$356.00 (1 filled call of 4 attempted)
- Concentration: one 4x QQQ long option on an overnight gap = 61% of the result — path, not skill
- Peak simultaneous defined worst case \$3,421.00 (3.42%), carried overnight; max drawdown \$33.15 on cycle-spaced samples, the overnight move not sampled; book flat

Public journal, vetoes, and originality

- Every cycle's proposal or no-trade result is journaled with its gate vector and rationale — rejections logged with the same fidelity as fills
- The declared variance split is the originality claim, not a marketed edge
- 29 vetoed candidates and 24 no-trade cycles are visible on the dashboard as of 2026-09-03T20:00:14.787Z

Failure drills, two defects found, and limitations

- Failure-path tests cover reconciliation, kill-switch, watchdog, expiry
- **Defect 1:** on flatten day the runner could not price the structures expiring next session (quote window starts at `EXPIRY_MIN_SESSIONS`); the refusal reached only a discarded report, so no close was submitted
- **Response:** owner stood the writer down, the certified dead-man watchdog took over — `HALT_WATCHDOG_TAKEOVER`, all three structures closed and filled within a second, book flat at the deadline
- **Defect 2:** that watchdog's wrapper had killed the CLI on its first stderr line since arming — inert for a day, fixed at 17:41 CEST, an hour before the takeover
- **API limit:** no conditional submit at Alpaca; the pre-submit re-fetch is the declared linearization point, a manual mutation halts the next cycle
- Paper only; two sessions of paper P&L cannot prove edge — no alpha claim

Demo, repository, and post-hackathon path

- Demo: <https://glass-box-trading.vercel.app>
- Repository: <https://github.com/fradzano/glass-box-trading>
- Journal revision at cutoff: sha256:7b82959a344a7c7e
- Post-hackathon: continue paper operation, extend the gate catalog and sleeve set before any live-account discussion